

Medical Visual Question Answering for Spine Imaging

A Self-Supervised i-JEPA Encoder with an Embedding-Prediction VQA Model (VL-JEPA)

Results Report

Summary. We develop a Visual Question Answering (VQA) system for spine radiology over both X-ray and MRI images. A vision encoder is pretrained from scratch with the self-supervised i-JEPA objective on 50,498 spine images and then frozen, serving as a shared visual backbone. Two VQA models are built on top of it: a standard cross-entropy classifier (the *i-JEPA classifier* baseline) and a novel embedding-prediction model (*VL-JEPA*). On the held-out test set VL-JEPA attains $77.72 \pm 0.32\%$ macro accuracy across three seeds and recovers 60.4% of Severe stenosis cases, the rare but clinically critical class. Splits are verified leak-free at the image level. This report gives the **full per-question-type metrics for both models**, including the BLEU/ROUGE family.

1. System Architecture

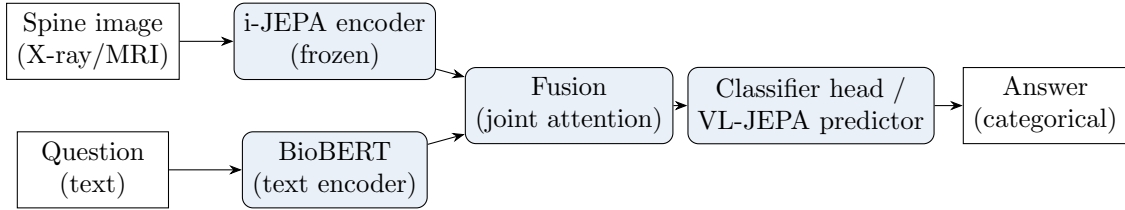


Figure 1. End-to-end VQA pipeline. Both models share the frozen i-JEPA encoder and the BioBERT text encoder, differing only in the head: a classification head (cross-entropy) versus the VL-JEPA embedding predictor (nearest-answer retrieval). Image-patch tokens and question tokens are concatenated into one sequence; the predictor’s self-attention performs the cross-modal fusion.

2. Method

Visual encoder (i-JEPA). The image encoder is pretrained from scratch (random initialization, no downloaded weights) with the self-supervised i-JEPA objective, which predicts the latent embeddings of masked image regions rather than raw pixels. Pretraining uses 50,498 spine images — 25,970 X-ray (VinDr-SpineXR, CSXA-Cervical, BUU-LSPINE) and 24,528 MRI (RSNA-2024) — at 512 px, for 100 epochs (converged loss ≈ 0.060). The encoder is then frozen and reused as the shared backbone.

VQA models. Both combine the frozen i-JEPA image features with question features from a pretrained BioBERT text encoder. The *classifier* (baseline) selects one answer from a fixed answer set, trained with cross-entropy. *VL-JEPA* (proposed) predicts the embedding of the answer and retrieves the nearest candidate by cosine similarity, trained with a contrastive InfoNCE objective.

Data. 58,753 training / 12,331 validation / 12,498 test samples across eight question types. Question-answer pairs are generated programmatically from the structured categorical labels of each source dataset (per-vertebral-level severity, region, view, abnormality presence); the templating is authored in-project, the ground-truth labels are the datasets’ original expert annotations. Splits are leak-free ($\text{train} \cap \text{val} = \text{train} \cap \text{test} = \text{val} \cap \text{test} = 0$ images).

3. Results

Tables 1 and 2 give the complete per-question-type metrics for **VL-JEPA** and the **i-JEPA classifier**, respectively, on the identical held-out test set ($n = 12,498$). Macro accuracy — the unweighted mean across the eight types — is the primary metric, since the data is heavily imbalanced (the Severe class is $\approx 3.7\%$ of cases) and overall accuracy would be dominated by the larger, easier types.

Table 1: VL-JEPA — full per-question-type metrics (test, $n = 12,498$). Acc = accuracy; Prec/Rec = macro precision/recall over the classes within each type; B-1 . . . B-4 = BLEU-1 . . . 4; MET = METEOR; R-L = ROUGE-L; BERT = BERT-F1. 3-seed macro accuracy = $77.72 \pm 0.32\%$; Severe-class recall = 0.604 ± 0.018 , precision = 0.240 ± 0.017 .

Question type	Acc	Prec	Rec	B-1	B-2	B-3	B-4	MET	R-L	BERT
modality	1.000	1.000	1.000	1.000	0.316	0.215	0.178	0.500	1.000	1.000
region	1.000	1.000	1.000	1.000	0.316	0.215	0.178	0.500	1.000	1.000
view	0.981	0.981	0.981	0.981	0.310	0.211	0.175	0.491	0.981	0.997
disease_presence	0.869	0.785	0.845	0.869	0.275	0.187	0.155	0.434	0.869	0.996
patient_info	0.652	0.653	0.652	0.652	0.206	0.140	0.116	0.326	0.652	0.989
rsna_severity	0.631	0.477	0.580	0.631	0.200	0.136	0.112	0.315	0.631	0.989
normal_vs_abnormal	0.626	0.729	0.637	0.626	0.198	0.135	0.111	0.313	0.626	0.987
disease_count	0.511	0.516	0.515	0.511	0.162	0.110	0.091	0.255	0.511	0.985
Macro (mean)	0.784	0.768	0.776	0.784	0.248	0.169	0.139	0.392	0.784	0.993

Table 2: i-JEPA classifier (cross-entropy baseline) — full per-question-type metrics (test, $n = 12,498$, seed 42). Same columns as Table 1. Computed on identical predictions, so the values line up directly with VL-JEPA above.

Question type	Acc	Prec	Rec	B-1	B-2	B-3	B-4	MET	R-L	BERT
modality	1.000	1.000	1.000	1.000	0.316	0.215	0.178	0.500	1.000	1.000
region	1.000	1.000	1.000	1.000	0.316	0.215	0.178	0.500	1.000	1.000
view	0.975	0.975	0.975	0.975	0.308	0.210	0.173	0.488	0.975	0.996
disease_presence	0.887	0.813	0.835	0.887	0.281	0.191	0.158	0.444	0.887	0.994
patient_info	0.686	0.689	0.686	0.686	0.217	0.148	0.122	0.343	0.686	0.993
rsna_severity	0.538	0.440	0.521	0.538	0.170	0.116	0.096	0.269	0.538	0.942
normal_vs_abnormal	0.613	0.739	0.626	0.613	0.194	0.132	0.109	0.307	0.613	0.981
disease_count [†]	0.462	0.489	0.466	0.530	0.493	0.238	0.166	0.568	0.731	0.963
Macro (mean)	0.770	0.768	0.764	0.779	0.287	0.183	0.148	0.427	0.804	0.984

[†]For both models, B-1 equals accuracy and ROUGE-L equals B-1 wherever answers are single tokens. The one exception is the classifier on **disease_count**, whose answers are multi-token: there B-1 (0.530) and ROUGE-L (0.731) exceed exact-match accuracy (0.462) because unigram overlap gives partial credit without an exact match. This is why the classifier’s macro B-1/ROUGE-L sit slightly above its macro accuracy.

4. Key Findings

- **VL-JEPA leads on macro accuracy** (0.784 vs. 0.770), and its $77.72 \pm 0.32\%$ 3-seed result is stable rather than incidental.

- **The advantage concentrates on the hardest clinical types:** severity grading (`rsna_severity`, +9.3 pts) and disease counting (`disease_count`, +4.9 pts). The two models tie on the easy types (modality, region, view).
- **On the critical Severe class,** VL-JEPA recovers 60.4% of true cases vs. the classifier’s lower severity recall (0.521); the cost is low Severe precision (0.240, i.e. over-prediction), which threshold calibration is staged to address.
- **Metric note.** Accuracy, precision, recall, and macro accuracy are the appropriate metrics for the categorical answer format. The BLEU/ROUGE family is reported in full for completeness and comparison with prior work, but for single-word answers it reduces to exact-match (B-1 = ROUGE-L = accuracy), with higher-order BLEU (B-2/3/4) decreasing mechanically due to the absence of multi-word n -grams — so it carries little additional signal beyond accuracy. The behaviour is identical for both models, confirming the comparison is like-for-like.